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$N = 250$

$B = 1,25$

$\mu r = 2500$

$r_i = 8\,cm \quad r_o = 12\,cm \quad r_{avg} = \frac{8+12}{2} = 0,1\,m$

a) $B = \mu H$

$\mu = \mu_0 \mu r$

$NI = Hl$

$\mu = (4\pi \times 10^{-7})(2500) \approx 3,14 \times 10^{-3}$

$\frac{12-8}{2} = 2cm$ ~~0,002m~~ $0,02m$

$A = \pi r^2 = \pi (0,02)^2 \approx 1,256 \times 10^{-3}$

$NI = Hl = 398 \times 0,628 \approx 250$

$I = 1A$

b) Magnetic flux

$\Phi = BA = 1,25 \times 1,256 \times 10^{-3} \approx 1,57 \times 10^{-3} Wb$ $1,57\,mWb$ Mag. flux

c) Air gap current

$l_g = 0,01\,m \qquad l_c = 0,062\,8$

Total ~~MMF~~ MMF =

$NI = H_c l_c + H_g l_g$

Φ should be constant

$B = \frac{\Phi}{A} = 1,25\,T$

for celik $\Rightarrow H_c = \frac{B}{\mu_0 \mu r} \approx 398\,A$

$NI = (398)(0,628) + (1,0 \times 10^6)(0,01) \approx 250 + 9950 \approx 10,200$

40,8A New current $I = 41\,A$

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d) Magnetic field intensity (H)

Gelik : $H_c = 338\ A/m$

Air space : $H_g = 1.0 \times 10^6\ A/m$

e) Inductance , (L)

$L = \frac{N\Phi}{I}$ firstly (NO air gap) $\Rightarrow$ $L = \frac{250 \times 1,57 \times 10^{-3}}{1} \cong 0,392\ H$

with air gap $\Rightarrow$ $L = \frac{250 \times 1,57 \times 10^{-3}}{40,8} \cong 9,6 \times 10^{-3}\ H$

Inductance is excessive decrease

f) Stored Energy in Inductor

$W = \frac{1}{2} L I^2$

for Starting $\Rightarrow$ $W = 0,5 \times 0,392 \times (1)^2 = 0,196\ J$

with air gap $\Rightarrow$ $W = 0,5 \times 0,0096 \times (40,8)^2 \cong 8,0\ J$

Energy is excessive increase because air gap provide the more big area for storage the magnetic energy.

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